

Line Following Robot – Algorithm

1. Power On State

- Robot is idle.
- Motors are OFF.
- Awaiting **first reset button press**.

2. First Reset Press – Calibration Phase

➤ A. Calibrate White Surface

- Robot stays on white surface.
- `qtr.calibrate()` is called 200 times.
- Records each sensor's minimum values.

➤ B. Move to Black Line

- Robot drives forward ~15 cm (`delay(600)`).

➤ C. Calibrate Black Line

- Again, `qtr.calibrate()` is called 200 times.
- Records each sensor's maximum values.

➤ D. Threshold Calculation

`threshold[i] = (min[i] + max[i]) / 2;`

- Now the robot can distinguish black/white on each sensor.

3. Second Reset Press – Start Operation

- Robot sets started = true.
- Enters main loop `lineFollow()`.

4. `lineFollow()` Execution

➤ A. Read Sensor Values

```
qtr.read(sensorValues);
```

➤ B. Convert to Binary

```
binarySensors[i] = (sensorValues[i] > threshold[i]) ? 1 : 0;
```

5. Finish Zone Detection

If:

`(centerHigh && outerHigh) || (centerLow && outerLow)`

Means:

- Center sensors detect line
- Outer sensors suddenly detect the same line color

✅ Robot stops and buzzer plays → Finish Reached

6. No Line Found Detection

If:

`(centerHigh && outerLow) || (centerLow && outerHigh)`

Robot will:

1. Move forward 85 ms (to cross potential 10 cm gap)
2. If still no line → spin left until any sensor detects line again

7. Junction Detection

➤ A. T-Junction

- `sensor[0] && sensor[7]` detect line
- Followed by **loss of center line**
- Detected as T-junction

► B. Y-Junction

- `sensor[2] && sensor[5]` detect same line color
- Center sensors are ignored

✓ Robot turns based on `turnRightAtJunction` until line is reacquired

8. Angle Turn Detection (90°)

► Left Turn

- If sensors 0,1,2 detect the line → turn left until center finds line

► Right Turn

- If sensors 5,6,7 detect the line → turn right until center finds line

9. Cave Detection

Triggered **only if no line is found**.

If sonar finds wall 2–50 cm away:

- If walls on both sides → follow center path
- If one wall → maintain 12 cm distance using:

`correction = error * 3;`

10. Regular Line Following

A. Determine Mode:

```
useWhiteLine = (centerLow && outerHigh);
```

B. Read Position:

```
pos = qtr.readLineWhite() or qtr.readLineBlack();
```

C. Steering Logic:

pos Range	Action
< 2000	Turn left
2000-5000	Go straight
> 5000	Turn right

11. Reset During Operation

- If reset is pressed again:
 - Operation restarts
 - Robot skips calibration
 - Continues with stored thresholds